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Deglycosylation induces a novel distal conformation in the mannose receptor CD206

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Abstract

Deglycosylation can disrupt protein structure and dynamics in ways that compromise function. In this study, we examine the structural and functional consequences of N-glycan removal in the mannose receptor CD206, a multi-domain lectin involved in innate immunity. Using atomistic molecular dynamics simulations and AlphaFold-predicted full-length models, we provide the first comprehensive structural and dynamical characterization of CD206, which remains unresolved by experimental methods. Our simulations reveal that deglycosylation enables access to a previously undescribed concave conformational state in the distal lectin domains CTLD7–8, in contrast to the glycosylated form, which remains restricted to a convex arrangement. Principal component and normal mode analyses show that this conformational switch is tightly linked to the presence of glycosylation in certain residues, which constrains receptor flexibility and reduces conformational sampling. While both conformations share an extended global architecture, they differ in curvature, potentially influencing ligand accessibility. Functional assays with α-D-mannopyranoside (MMA), a natural-occurring ligand of CD206, and the CD206-

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binding tumor-homing peptide mUNO show that deglycosylation weakens canonical ligand retention and exposes new interaction sites, including regions normally shielded by glycans. These findings suggest that altered glycosylation, as seen in pathological conditions like cancer, can reshape the receptor's conformational and binding landscape, ultimately influencing immune recognition and offering new opportunities for therapeutic targeting.

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Introduction

The mannose receptor (MR), also known as CD206, is a transmembrane C-type lectin expressed in macrophages, immature dendritic cells, and endothelial cells. It plays a central role in innate immunity by recognizing and internalizing a wide range of ligands—including sulfated sugars, glycoconjugates, and collagen—through multiple specialized domains: a cysteine-rich (CysR) domain, a fibronectin type II (FNII) domain, and eight C-type lectin-like domains (CTLDs). Beyond its membrane-bound form, CD206 can also be found in circulation as a soluble isoform (sMR) (1**),** which has been correlated with the severity of various inflammatory and fibrotic diseases, making it a potential biomarker and therapeutic target.

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A key feature of CD206 is its N-glycosylation, which has been shown to modulate ligand binding, surface expression, and immune interactions (2-5**)**. Previous studies identified glycan signatures that vary by tissue (6, 7), and specific glycosylation patterns have been shown to affect the activity of certain CTLDs, particularly CTLD4 and CTLD5, which are directly involved in mannose recognition (**8**). Interestingly, many glycosylation sites are located in domains that do not directly contact ligands, suggesting a potential indirect or allosteric regulatory role (9**).** It is proposed that N-glycans can act as switches that mask or expose self-ligands depending on the cellular context.

More broadly, glycosylation is known to influence protein dynamics and stability. Computational studies across diverse systems (10,11) have shown that N-glycans can reduce local flexibility and promote structural compactness, often without altering the fundamental fold of the protein. In addition, glycosylation plays essential

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roles in protein folding, trafficking, cell adhesion, immune regulation, and signal transduction. Aberrant glycosylation is strongly associated with cancer progression, immune evasion, hematological malignancies, autoimmune and neurodegenerative diseases. Although it is a highly active area of research, some findings remain inconclusive or even contradictory, reflecting the complexity and heterogeneity of glycan structures. Recent advances highlight its potential as a source of diagnostic and prognostic biomarkers and as a therapeutic target, including inhibitors, engineered antibodies, and vaccines, yet significant challenges remain (2). In the context of CD206, such dynamic effects are particularly relevant given its modular, extended architecture and the functional importance of its inter-domain flexibility.

Despite the central role of glycosylation in CD206 biology, the full-length structure of the receptor remains unresolved experimentally, and the dynamic behavior of its distal CTLD domains—particularly CTLD7 and CTLD8—is poorly understood. This limits our ability to interpret how conformational changes might impact ligand recognition, especially in pathological contexts such as cancer, where glycosylation patterns are frequently altered (12,13).

In this study, we use atomistic molecular dynamics simulations to investigate the structural and dynamic consequences of N-glycan removal in CD206. Starting from full-length models generated by AlphaFold, we report the first atomistic view of the complete receptor and identify a novel conformational state in CTLD7–8, accessible only in the de-glycosylated form. We further show that this structural plasticity modulates ligand-binding behavior, offering a mechanistic explanation for the differential recognition of the tumor-homing peptide mUNO. mUNO is a synthetic peptide that selectively targets CD206 on tumor-associated macrophages (TAMs), a cell population with key immunosuppressive roles in the tumor microenvironment. mUNO has emerged as a promising targeting tool in cancer immunotherapy (14). Our simulations suggest that glycosylation-dependent conformational shifts in CD206 alter the accessibility and geometry of the mUNO-binding pocket, thereby providing a molecular rationale for its selective recognition. Taken together, our findings reveal that glycosylation exerts long-range regulatory effects on CD206 function and underscore the importance of distal domain dynamics in shaping ligand interactions and, ultimately, in modulating immune responses.

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Results

Structural Consequences of N-Glycan Removal on the CD206 Receptor

To investigate the structural consequences of N-glycan removal, we performed atomistic molecular dynamics simulations of glycosylated (G) and non-glycosylated (NG) full-length CD206 starting from AlphaFold-predicted structures. These models include all known extracellular domains—CysR, FNII, and CTLD1–8 **(Supplementary Figures S1,S2)**—allowing us to capture inter-domain interactions often missing from partial or fragmented models.

Before examining the large-scale rearrangements of the receptor, we first evaluated the structural stability and residue-level flexibility of both G and NG CD206 across three independent simulations. The average RMSD profiles confirmed that all trajectories remained stable throughout the simulations, with each domain preserving its fold (**Supplementary Figure S3B-C**). Although some NG replicas display large RMSD values, inspection of domain-resolved RMSD and structural superpositions (Figure S3D) shows that these deviations arise from rigid body repositioning of domains rather than from unfolding. Domain-resolved RMSD analysis further indicated that the largest deviations occur in the C-terminal CTLD7–8 region, consistent with the inter-domain motions described below (**Supplementary Figure S3A).** Some domains display bimodal RMSD distributions, suggesting the presence of alternative local conformations. Complementarily, the per-residue RMSF analysis revealed that glycan removal led to an overall increase in local flexibility, particularly in the CTLD7–8, and to a lesser extent, the N-terminal CysR domain **(Supplementary Figure S4)**. To compare this effect across replicas, Figure S4 also reports the standard deviations of the RMSF profiles for each system. The difference profile ΔRMSF highlights a marked mobility gain in CTLD6–8 upon deglycosylation.

Analysis of the trajectories revealed that both G and NG systems predominantly maintain an extended architecture consistent with the conformation observed experimentally at physiological pH (15). However, striking differences were observed in the distal region encompassing CTLD7 and CTLD8.

To objectively characterize these conformational differences observed, the trajectory ensembles were clustered using the *k*-means algorithm implemented in *cpptraj* (see Methods), based on the Cα RMSD matrix of residues 1–1390. This analysis revealed two dominant conformational groups that correspond to distinct arrangements of the distal

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lectin domains: one displaying a convex configuration, in which CTLD7–8 are oriented outward from the receptor's core axis toward the extracellular space, and a second, previously undescribed concave arrangement, characterized by the inward bending of CTLD7–8 toward the intracellular side (**Fig. 1**). To quantify these conformational transitions, we defined an inter-domain angle formed by three representative residues from CTLD3, CTLD7, and CTLD8, which captures the degree of bending or extension of the distal lectin region. As shown in **Supplementary Figure S5**, the G system—favoring the convex conformation—exhibits smaller angles (median ≈ 85°), corresponding to an outward-facing distal arrangement. In contrast, the NG system samples larger angles (median ≈ 145°), consistent with the inward-bent, concave geometry.

This conformational switch was only observed in the NG system, which sampled both convex and concave states, whereas the G system remained confined to the convex conformation. These findings suggest that deglycosylation relieves conformational constraints, enabling access to alternative states that may be functionally relevant.

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Principal Component Analysis

To further investigate the origin of the conformational switch, we performed principal component analysis (PCA) on the concatenated trajectories of G and NG systems. Following the approach of (16), we built a single covariance matrix from the combined atomic fluctuations of both systems, enabling direct comparison of motions in a shared conformational space. Since PCA eigenvalues represent the mean square fluctuations of each principal mode, and their sum equals the total atomic mean square fluctuations, we focused on the first two eigenvectors, which together account for 88% of total variance. This defines a two-dimensional essential subspace described by PC1 and PC2. Projection onto these components (**Fig. 2A**) reveals a clear separation of conformational states, with NG sampling two major basins—one overlapping with G and another unique to NG. Coloring the projection by simulation (**Fig. 2B**) shows that G populates only the convex state, whereas NG samples both convex and concave states.

To better understand the structural nature of this transition, we visualized the first eigenvector (PC1) as a vector field over the protein structure (**Fig. 2C**). The dominant motion aligns with the rearrangement of the distal CTLD7–8 region, confirming that

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PC1 captures the conformational switch between convex and concave states. The remaining 12% fluctuations are mainly described by PC3 and PC4. They were found to correspond to a breathing-like motion involving the expansion and contraction of the distal lectin region. These modes do not distinguish between conformational basins but reflect local flexibility within each state. Notably, the NG system samples broader regions along both PC2 and PC3, suggesting enhanced conformational plasticity in the absence of N-glycans.

Functional Consequences of CD206 Conformational Plasticity

To assess how deglycosylation-driven structural changes influence ligand-binding properties, we analyzed the interaction of two ligands with glycosylated (G) and non-glycosylated (NG) receptor systems: methyl α-D-mannopyranoside (MMA), a canonical ligand for CTLD4 (17**)**, and mUNO, a CD206-binding peptide that targets CD206+ tumor associated macrophages in primary and secondary tumors (18,19). These ligands differ in size, chemical nature, and binding mode—MMA binds to a defined, calcium-coordinated pocket of CTLD4, whereas mUNO is predicted to engage broader and more flexible receptor surfaces. We directly compared G and NG systems to determine how the absence of N-glycans—and the resulting structural and dynamical changes—affects ligand accessibility and binding stability.

For MMA, we focused on isolating the specific effects of N-glycan removal while controlling for large-scale conformational variability. To achieve this, we simulated ligand binding starting from the convex conformation—the dominant state shared between both G and NG ensembles. This design ensured that any observed differences in ligand behavior arose directly from deglycosylation, without the confounding influence of distal domain rearrangements. Although the CTLD4 domain, which harbors the MMA binding pocket, shows minimal structural deviation between G and NG systems, deglycosylation led to marked changes in ligand retention and contact patterns. In the NG system, MMA exhibited altered binding poses and reduced stability within the pocket, suggesting that glycans can modulate ligand recognition at structurally conserved sites through long-range dynamic coupling rather than direct steric effects. The dissociation process was analyzed from the MMA-CD206 trajectory, generated for each glycosylation state. In the NG system, this single trajectory captures a complete unbinding event, allowing us to follow the sequential disruption of peptide–receptor contacts and identify the

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residues involved in the release pathway. In the glycosylated system (G), MMA remained stably bound within the canonical mannose-binding site of CTLD4 throughout the 1 μs simulation, maintaining key interactions such as calcium coordination and hydrogen bonding with residues Asn721, Glu733, and Tyr748 (**Fig. 3A**). The ligand showed minimal displacement and remained buried in a compact binding pocket. In contrast, in the non-glycosylated system (NG), MMA rapidly dissociated from the canonical site and migrated across the receptor surface, adopting multiple transient poses with reduced burial and contact frequency (**Fig. 3B**). Contact analysis revealed the appearance of new, superficial binding interactions (e.g., with Ile755 and Glu800), and hydrogen bond occupancy dropped markedly relative to G (**Fig. 3C**). Consistent with this behavior, residue-level flexibility analysis revealed increased mobility in the CTLD4–6 segment of the NG receptor (Figure S4), supporting the dynamical origin of the observed loss of peptide retention. These findings suggest that N-glycans indirectly stabilize ligand retention at CTLD4, possibly by modulating the local geometry or dynamics of the binding pocket, even though the glycans are located distally from the binding site.

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-mUNO

We next examined the interaction of the tumor-homing peptide mUNO with CD206 to directly assess how deglycosylation influences its binding behavior. Given that the previously reported binding pose (19) was characterized using only a partial receptor model (which comprised the domains CysR-FNII-CTLD1-2), we now placed four mUNO peptides at random positions across the full-length G and NG receptor models. Molecular dynamics simulations were then performed to evaluate how glycan removal affects peptide recognition, site accessibility, and interaction persistence over time.

Visual inspection of the trajectories, combined with volumetric density mapping and clustering analyses, revealed that mUNO explores distinct regions of the receptor surface in G and NG systems. In the NG simulation, some of the most frequently visited binding sites overlapped with regions that are sterically blocked by glycans in the G system, suggesting that glycan removal opens new interaction surfaces.

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Although the concave distal conformation was not sampled in the NG+mUNO trajectory, the CTLD7–8 region exhibited enhanced flexibility, with fluctuations along the dominant PCA1 mode. This mobility was sufficient to define two major conformational clusters, highlighting increased dynamical variability in the absence of N-glycans, even without a full conformational switch.

To quantify mUNO binding, we clustered ligand positions over the course of the simulations. In the G system, four major binding poses were identified (**Fig. 4A, bottom**), each sampled for 30–50% of the simulation time. The most frequently sampled pose (pose 1) localized to CTLD5 (residues K818, E819, T820, N823, and A826), while pose 2 recapitulated a previously reported interaction site (19**)**, reinforcing its biological relevance. Pose occupancies were relatively balanced across poses 1–4, with no pose dropping below ~30% (**Fig. 4B)**. In the NG system, binding was more diffuse and dynamic, with only three primary poses consistently sampled. The most frequent pose occurred at the CTLD3–CTLD4 interface (residues Q571, T572, Q577, W578, and T579). Additionally, a unique NG-only pose (pose 5) was found in a region overlapping with two glycan positions in the G system (**Fig. 4A, top**), highlighting how deglycosylation increases accessibility to previously occluded surfaces.

Notably, the total number of mUNO contact events was substantially reduced in NG compared to G (**Fig. 4B**). This reduction, combined with the increased dispersion of binding poses and reduced recurrence of specific interactions, points to a significantly weaker and more transient interaction profile in the NG system. These findings indicate that deglycosylation may lower the affinity to the receptor by increasing the conformational entropy of the binding interface.

Discussion

The effects of N-glycosylation on CD206 extend beyond local structural modifications at the glycan attachment sites, influencing the receptor's conformational landscape, interdomain communication, and ligand-binding behavior. Our simulations demonstrate that glycan removal induces structural and dynamic changes in distal regions of the receptor, including those not directly glycosylated and not traditionally recognized as binding sites.

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A quantitative analysis of the dynamic cross-correlation matrices (**Fig. 5**) confirms that the most significant glycosylation-dependent differences occur in the distal region of the receptor. In particular, the CTLD7 and CTLD8 domains exhibit systematically higher internal correlations in the non-glycosylated system. For CTLD7, both the mean and median correlation values increase in NG, together with a larger fraction of strongly correlated residue pairs ($|C| > 0.5$). This trend becomes even more pronounced in CTLD8, where NG shows a substantial rise in both average correlation and the proportion of high-correlation pairs. When CTLD7 and CTLD8 are considered together as a distal block, NG still displays stronger dynamical coupling than the glycosylated receptor. These results indicate that deglycosylation enhances concerted motions specifically within the distal lectin modules, providing a dynamical basis for the conformational transition captured in the simulations.

Beyond these distal effects, the DCCM also reveals long-range communication patterns across the modular architecture of CD206. Correlated fluctuations between the CysR domain and the CTLD4–6 region—despite being separated by more than 400 residues—suggest that the receptor's spatial organization enables propagation of dynamical influence between distant modules. Within this region, the strongest correlations involve the CysR domain and a segment in CTLD4 that includes the mannose-binding pocket, the only lectin module capable of independent carbohydrate recognition (8). The proximity of this dynamically coupled region to key binding residues (E725, E733, N727, Y729, N747) highlights the potential for glycosylation-dependent modulation of function through indirect, architecture-mediated communication.

A second region of interest is the CTLD1–2 module, which also contains glycosylation sites and exhibits dynamic coupling to the distal CTLD7–8 domains. This is particularly relevant given our identification of a glycosylation-dependent conformational switch in CTLD7–8, which reshapes the distal architecture of CD206. Although the mUNO peptide does not bind directly in this region, its observed differences in binding patterns and affinities between glycosylated and non-glycosylated systems may stem from these long-range dynamical effects.

In this context, the lower binding affinity and altered pose distribution of mUNO in the NG system (**Fig. 4**) are especially noteworthy. These results suggest that N-glycans contribute to receptor preorganization, stabilizing conformations that are more favorable for high-affinity ligand engagement. This allosteric effect may contribute to the observed preferential targeting of mUNO to CD206 in cancerous

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tissues (where N-glycosylation has been reported to be increased (20)) over CD206 in the liver (21-22). The overlap of mUNO binding sites in NG with glycan-occupied regions in G further reinforces this idea, highlighting the competitive or occlusive role of glycans in modulating surface accessibility.

Overall, our findings illustrate that glycosylation plays a critical role in shaping the dynamic and functional landscape of CD206. Through a combination of direct steric effects and long-range allosteric coupling, N-glycans regulate the accessibility, flexibility, and ligand selectivity of the receptor. This underscores the importance of considering glycosylation not merely as a static structural feature, but as an active modulator of receptor function—potentially exploitable for therapeutic targeting or biomarker design.

In summary, our study provides a comprehensive view of how N-glycosylation shapes the structure, dynamics, and ligand-binding properties of the human macrophage mannose receptor CD206. Using all-atom molecular dynamics simulations of glycosylated and non-glycosylated full-length receptor models, we uncover a previously unreported conformational switch involving the distal lectin domains CTLD7 and CTLD8. The new conformation, which is sampled exclusively in the absence of glycans, highlights the role of glycosylation in restricting conformational freedom and stabilizing specific receptor states.

Functional analysis through ligand binding simulations revealed that glycosylation promotes stable interaction of canonical ligands such as methyl α-D-mannopyranoside (MMA), and modulates the accessibility and affinity landscape of the CD206-binding peptide mUNO. In the deglycosylated system, mUNO samples different and more superficial binding sites with reduced persistence, suggesting that glycans contribute not only to steric shielding but also to long-range preorganization of the receptor surface.

Inter-residue dynamic correlation analysis further supports this interpretation, revealing long-range coupling between glycosylated domains (such as CysR and CTLD1–2) and functionally important regions like CTLD4 and CTLD7–8. These allosteric connections suggest that N-glycans can influence ligand binding through both direct and indirect mechanisms, adding a new layer of regulatory complexity to CD206 function.

Altogether, our results emphasize that glycosylation is a key determinant of CD206 architecture and functionality, with implications for immune recognition, ligand

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specificity, and therapeutic targeting. Understanding these molecular principles may aid in the design of glyco-modulated ligands or in exploiting CD206 as a biomarker in pathologies where glycosylation is dysregulated.

Methods

CD206 models were developed using AlphaFold2 (23), loading CD206 from UNIPROT code (24) P22897. Models were prepared for atomistic simulations using tLeap from Amber Package (25), adding TIP3P water molecules (26) and Na+ and Cl- ions to form a 0.2 M solution. Models were validated using Molprobity (27) and assignment of appropriate protonation states were handled using PDB2PQR (28, 29). Interactions were modeled using ff19SB amber forcefield (30). Full electrostatic interaction energies were calculated using the particle mesh Ewald method (31). Molecular dynamics simulations were performed using the AMBER package (25). Simulation details are found in **Supplementary Table S1**.

To construct the G model, the NG minimized structure was uploaded to the GLYCAM server (32) for addition of N-linked glycans at the seven experimentally annotated consensus sites: N104, N344, N529, N926, N930, N1160, and N1205 (9). Each site was modeled with a single N-acetylglucosamine (GlcNAc) residue covalently linked to the asparagine side chain, representing the conserved core of N-linked glycosylation. This minimal representation captures the structural and electrostatic contribution of the glycosidic linkage while avoiding additional conformational complexity from higher-order glycan branches. The carbohydrate parameters were assigned using the GLYCAM06j force field, fully compatible with the ff19SB protein parameters.

Three independent replicas were simulated for both G and NG systems following the same protocol. Each system was first minimized for 5000 steps (500 steepest descent followed by conjugate gradient) with restraints of 100 kcal mol^{-1} $Å^{-2}$ on protein heavy atoms. The systems were then gradually heated from 10 to 298 K over 1 ns under constant volume (NVT) conditions, using a Langevin thermostat (collision frequency 2 ps^{-1}). This was followed by a 400 ps pressure equilibration (NPT) using a Berendsen barostat (τ_p = 1 ps, target pressure 1 atm), and a 10 ns unrestrained equilibration under NVT at 298 K.

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Production simulations were carried out in the NVT ensemble, using a 2 fs integration step and SHAKE constraints on bonds involving hydrogens. Nonbonded interactions were truncated at 10 Å, and periodic boundary conditions were applied throughout. Trajectories were saved every 10 ps for analysis.

For the glycosylated and non-glycosylated systems, each replica was extended to 1 μs, resulting in a total simulated time of 3 μs per system. Clustering of the trajectories was performed using both hierarchical agglomerative and k-means algorithms, classifying frames according to their pairwise RMSD. Principal components (PCs) were obtained by diagonalizing the covariance matrix of Cα atoms after alignment to the average structure. Eigenvector visualization was performed with the Normal Mode Wizard in VMD (33).

Normal-mode calculations were conducted using the Anisotropic Network Model (ANM) implemented in ProDy, considering Cα atoms from the minimized structure. The first 20 modes were analyzed to identify collective domain motions and deformation energies.

Simulations with MMA and mUNO were performed in the same way as described above. For MMA, we first superimposed the configuration reported in (17) (PDB 7JUB) to the full model and continued with the process described for the NG system. For mUNO, we placed 4 peptides at each of the 4 corners of the simulation box and allowed the system to relax. Each simulation was 1 μs long.

Geometrical properties, rmsd values and correlation matrix were calculated with cpptraj (34).

Spatial occupancy maps of the MMA peptide were generated using the Density Map tool in Chimera (35). In this procedure, each simulation frame contributes a voxel at the coordinates of the ligand heavy atoms, producing a three-dimensional density representation of all positions visited during the trajectory. The resulting map thus reflects the spatial persistence of the ligand throughout the simulation rather than a geometric cavity volume.

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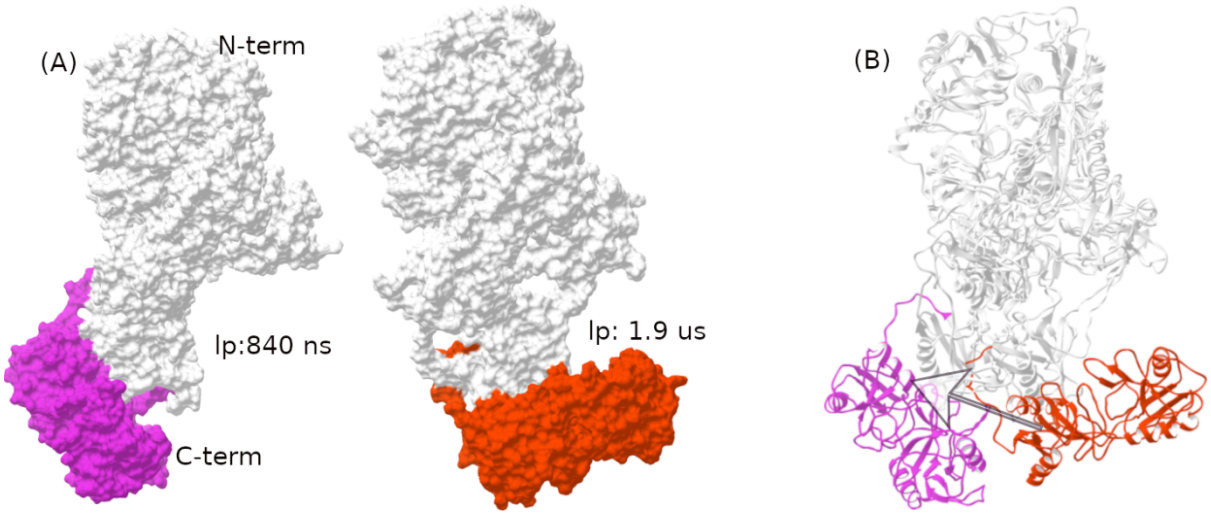


Figure 1. Distal conformational variability in CD206 revealed by molecular dynamics simulations.

(A) Surface representation of two representative conformations sampled during molecular dynamics simulations of CD206, corresponding to two distinct structural states of the distal lectin domains CTLD7 and CTLD8: a concave conformation (magenta) only observed for the NG ensemble, and a convex conformation (orange) present in both ensembles. Life permanence values for each state are indicated. **(B)** Ribbon representation of the same conformations, aligned by the central domains to highlight the difference in curvature of the CTLD7–8 region. This distal curvature variation defines a novel conformational switch modulated by deglycosylation in the mannose receptor CD206.

Figures were generated by the authors using VMD (version 1.9.3, https://www.ks.uiuc.edu/Research/vmd/)

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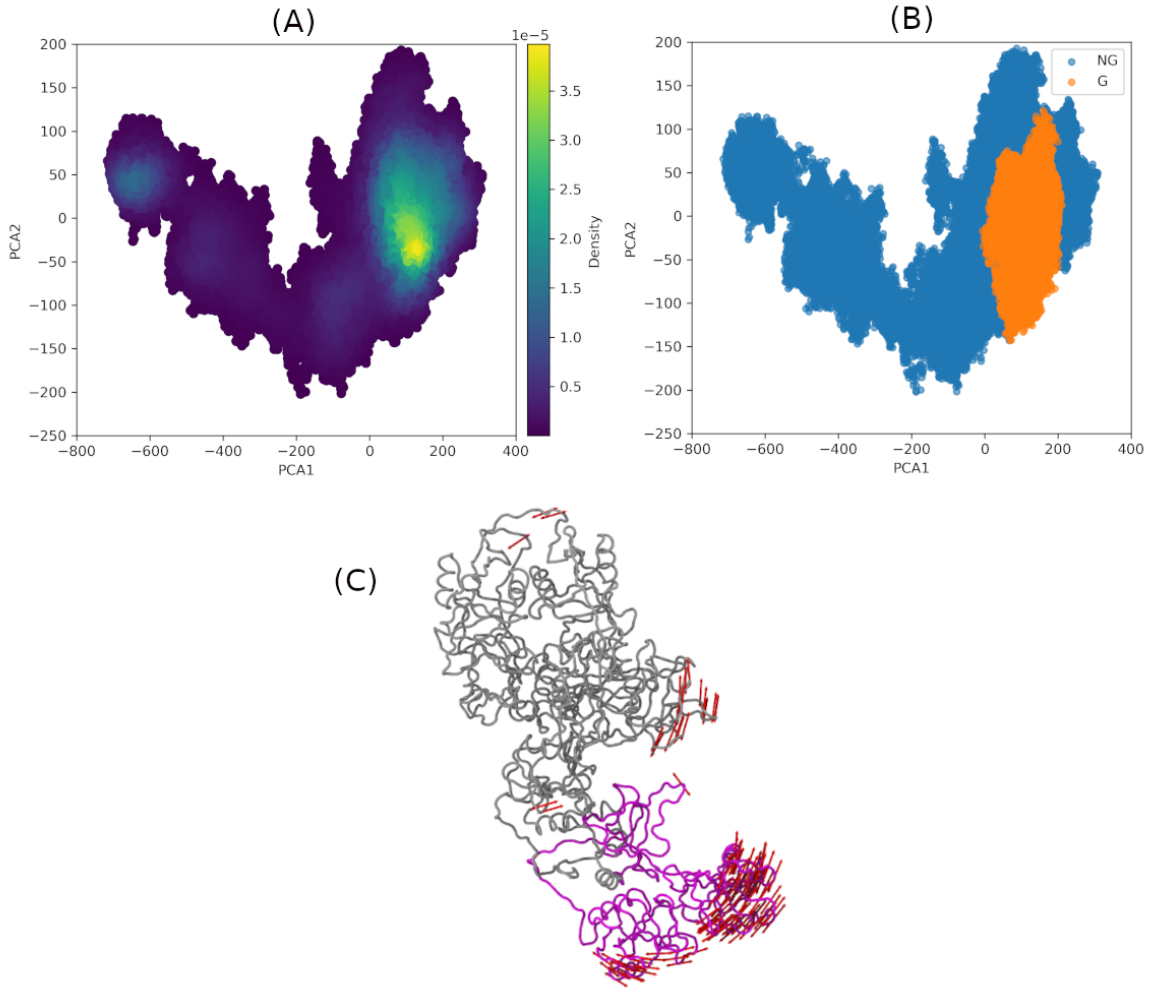


Figure 2: PCA and normal modes (A) Projection of all simulation frames from glycosylated (G) and non-glycosylated (NG) systems onto the first two principal components (PCA1 and PCA2), computed from a concatenated covariance matrix. The density plot reveals two major basins in the conformational landscape. **(B)** Same projection as in (A), colored by simulation. The G system (orange) samples a single basin corresponding to the convex state, while the NG system (light blue) explores both convex and concave conformations. **(C)** Vector representation of the first principal component (PCA1) displayed on the protein backbone, showing RMSD values greater than 2 Å. The dominant motion involves a collective rearrangement of the distal lectin domains CTLD7 and CTLD8, consistent with the conformational switch observed in the NG ensemble.

Panels A and B were generated by the authors using Python (version 3.9) with NumPy and Matplotlib. Panel (C) was generated using VMD (version 1.9.3, ttps://www.ks.uiuc.edu/Research/vmd/).

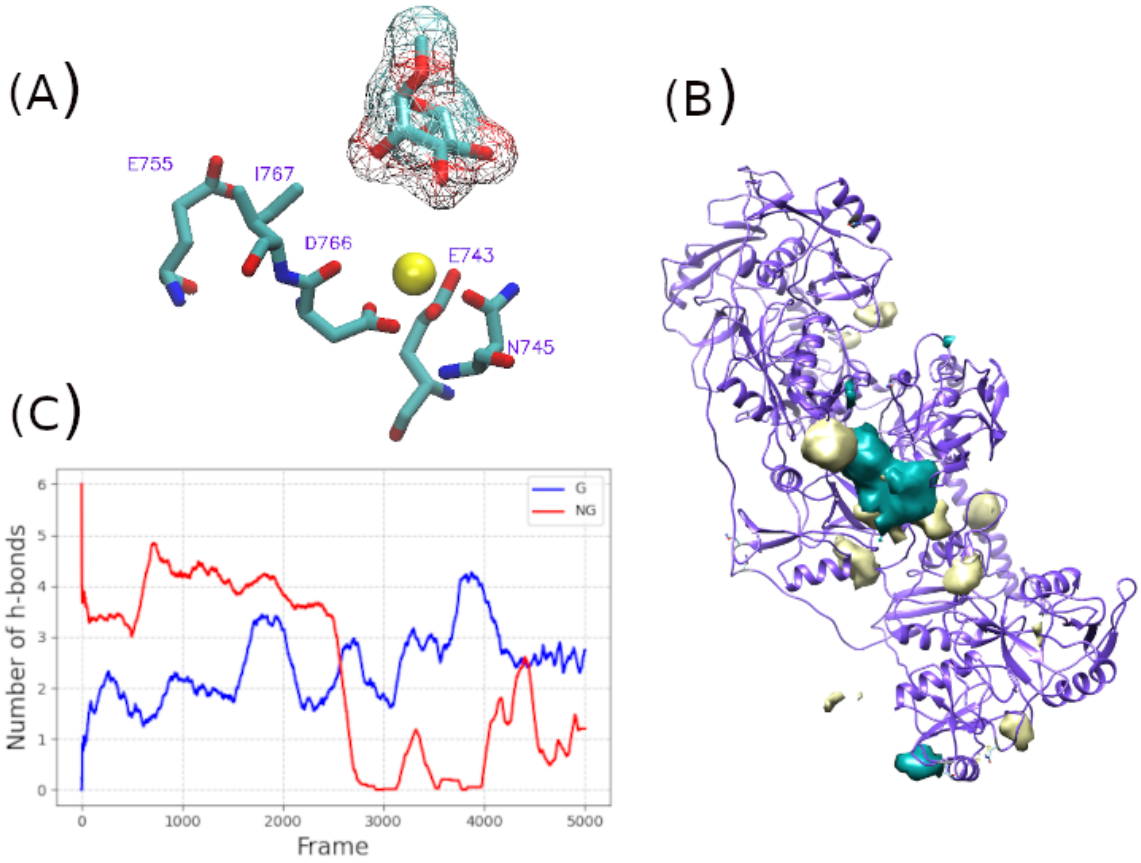
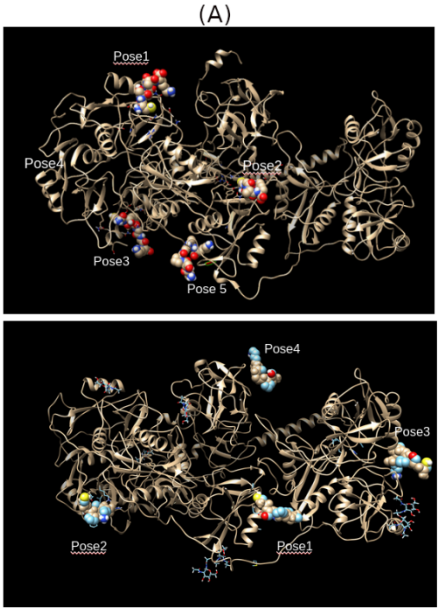


Figure 3. Glycosylation modulates the binding of MMA to CD206. (A) Structure of the canonical mannose-binding pocket in CTLD4, highlighting the bound MMA ligand in its initial position (licorice representation surrounded by a gray wireframe) and the coordinating calcium ion (yellow). **(B)** Spatial occupancy of the MMA peptide during molecular dynamics simulations in the glycosylated (G; cyan bubbles) and non-glycosylated (NG; light yellow bubbles) systems, showing the three-dimensional regions most frequently sampled by the ligand in each system. **(C)** Time evolution of the moving average of hydrogen bonds between MMA and receptor residues for the G (blue line) and NG (red line) systems.

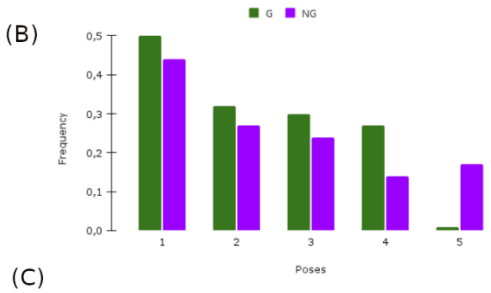
Panels (A) and (B) were generated by the authors using VMD (version 1.9.3, https://www.ks.uiuc.edu/Research/vmd/). Panel (C) was generated using Gnuplot (version 4.6, http://www.gnuplot.info/).

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(A)



(B)



(C)

	Pose 1	Pose 2	Pose3	Pose4	Pose 5
G	818-819-820823-826 (CTLD5)	420-421-465-467-476-480-482 (CTLD2)	1167-1168-1171-1173 (CTLD7)	1047-1049,73 (CysR-CTLD5,6)	
NG	571-572-577-578-579 (CTLD3-4)	903-910-911 (CysR-CTLD5)	324-326 (CTLD1)	367 (CTLD2)	810 (CTLD5)

Figure 4: Comparison of mUNO Binding Poses in G vs. NG Systems

(A-top) Structural mapping of the poses sampled in the non glycosylated system (NG), showing their spatial distribution across the receptor surface, including a fifth pose (Pose 5) located in a region normally occluded by N-glycans in the G system. **(A-bottom)** Corresponding poses identified in the glycosylated system (G),

(B) Relative occupation frequencies of each pose for G (green) and NG (magenta) systems, indicating weaker binding in the absence of glycans.

(C) Table listing the key residues involved in each pose, highlighting shifts in binding localization between systems.

Panel (A) was generated by the authors using VMD (version 1.9.3, https://www.ks.uiuc.edu/Research/vmd/). Panel (B) was generated using Gnuplot (version 4.6, http://www.gnuplot.info/).

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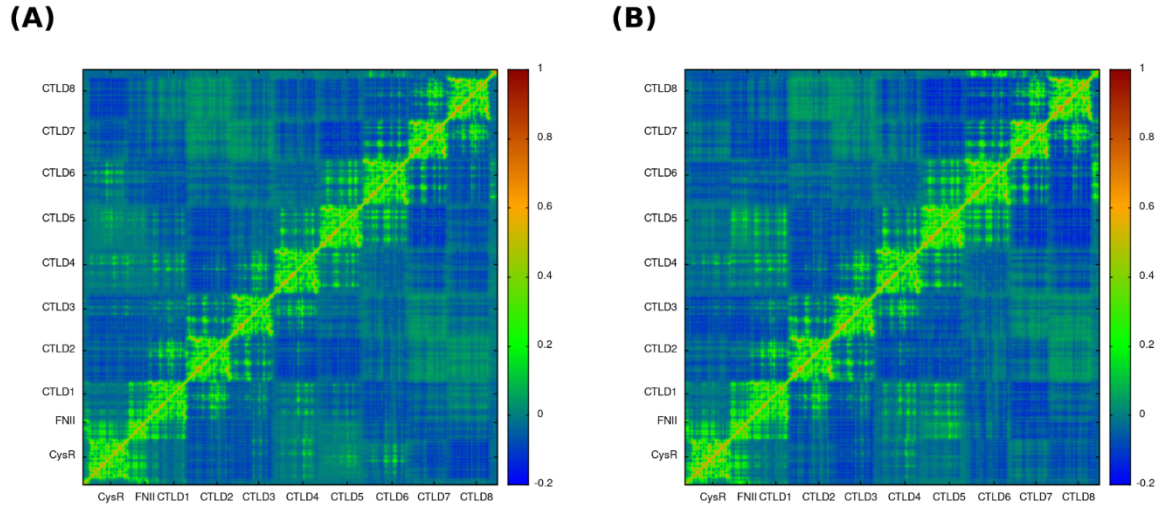


Figure 5. Dynamic cross-correlation matrices (DCCMs) for glycosylated (A) and non-glycosylated (B) CD206 derived from representative MD trajectories.

The matrices quantify the correlated positional fluctuations of Cα atoms along time for one representative trajectory of each glycosylation state. Positive values represent correlated motions, negative values anti-correlated motions, and values near zero indicate uncorrelated fluctuations. Both systems display strong intra-domain correlations characteristic of the modular architecture of CD206, while subtle differences in the distal CTLD7–8 region are apparent between glycosylation states, consistent with the localized dynamical effects quantified in our analysis.

Panels (A) and (B) were generated by the authors using Python with NumPy and Matplotlib (version 3.9).

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